UNIT 7 — ACIDS, BASES AND SALTS

1. INDICATORS

Indicators are substances that change colour in acidic or basic solutions and help identify whether a solution is acidic, basic (alkaline) or neutral.

Indicator	Acidic solution	Neutral solution	Basic / alkaline solution
Blue litmus	Turns red	Remains blue	Remains blue
Red litmus	Remains red	Remains red	Turns blue
Phenolphthalein	Colourless	Colourless	Pink
Methyl orange	Red	Orange	Yellow
Turmeric	Yellow	Yellow	Reddish-brown / red
Red cabbage extract	Red / pink	Purple	Blue / green (depending on strength)

Types of indicators

• Natural indicators are obtained from natural sources. Examples: litmus, turmeric, red cabbage extract and beetroot extract.

• Synthetic indicators are prepared chemically. Examples: phenolphthalein, methyl orange and methyl red.

• Universal indicator gives a range of colours corresponding to different pH values; it can indicate approximate acidity or alkalinity.

2. ACIDS

• Many acids taste sour, but never taste laboratory chemicals.

• Acids produce hydrogen ions (H■) in aqueous solution; more precisely, these exist as hydronium ions (H■O■) in water.

• Acids turn blue litmus red. They generally have pH below 7.

• Organic acids occur naturally in plants and animals; mineral acids are commonly obtained from inorganic sources.

Organic acid	Common source
Acetic acid	Vinegar
Ascorbic acid	Amla and citrus fruits (vitamin C)
Malic acid	Apples
Oxalic acid	Tomatoes and some leafy plants
Tartaric acid	Tamarind
Lactic acid	Curd / sour milk

Common mineral acids

Acid	Formula
Hydrochloric acid	HCl
Nitric acid	HNO■
Sulphuric acid	H■SO■
Phosphoric acid	H■PO■

Chemical properties of acids

• Acid + reactive metal → salt + hydrogen gas. Example: 2HCl + Zn → ZnCl■ + H■.

• Acid + carbonate / bicarbonate → salt + water + carbon dioxide. The CO■ can turn limewater milky.

• Acids react with bases to form salt and water (neutralisation).

• Not every metal reacts equally with every acid; the metal must be sufficiently reactive.

Uses of important acids

• Hydrochloric acid: used in chemical manufacture, metal cleaning/pickling and preparation of chloride salts. It is also present in gastric juice.

• Nitric acid: used in manufacture of fertilisers, dyes and certain industrial chemicals.

• Sulphuric acid: used in fertiliser manufacture, petroleum refining and lead-acid storage batteries; it is also an important industrial dehydrating agent.

• These concentrated acids are corrosive and must be handled only with proper laboratory safety.

3. BASES AND ALKALIS

• A base is a substance that neutralises an acid. Many bases produce hydroxide ions (OH■) in aqueous solution.

• Water-soluble bases are called alkalis. Therefore, every alkali is a base, but not every base is an alkali.

• Alkalis turn red litmus blue and generally have pH above 7.

• Examples of alkalis: sodium hydroxide (NaOH), potassium hydroxide (KOH), calcium hydroxide [Ca(OH)■] and aqueous ammonia (often represented as NH■OH in school texts).

• Examples of bases that are poorly soluble or insoluble in water include copper(II) hydroxide [Cu(OH)■], iron(III) hydroxide [Fe(OH)■] and aluminium hydroxide [Al(OH)■].

Uses of bases

• Sodium hydroxide: manufacture of soaps, detergents, paper and other chemicals.

• Calcium hydroxide (slaked lime): preparation of mortar and plaster, whitewashing, neutralising acidic soil and certain industrial processes.

• Magnesium hydroxide: used in some antacid preparations to neutralise excess stomach acid.

• Ammonia solution: used as a laboratory reagent and in some cleaning products.

Chemical properties of bases

• Bases react with acids to form salt and water.

• Alkalis produce OH■ ions in aqueous solution.

• Many bases feel slippery and taste bitter, but never touch or taste unknown chemicals to identify them.

4. DIFFERENCES BETWEEN ACIDS AND BASES

Property	Acids	Bases
Litmus	Blue litmus turns red.	Red litmus turns blue.
Phenolphthalein	Colourless.	Pink.
Methyl orange	Red.	Yellow.
Typical taste	Sour (never taste lab chemicals).	Bitter (never taste lab chemicals).
Ions in water	H■ / H■O■ ions.	OH■ ions (for common alkalis).
pH	Usually below 7.	Usually above 7.

5. NEUTRALISATION

Neutralisation is the reaction between an acid and a base to form salt and water. In a typical acid–alkali reaction: Acid + Base → Salt + Water.

Example: HCl + NaOH → NaCl + H■O.

Applications of neutralisation

• Antacids contain mild basic substances that neutralise excess acid in the stomach. Examples include magnesium hydroxide and some preparations containing sodium hydrogencarbonate.

• Farmers may add lime (such as calcium carbonate or calcium hydroxide) to acidic soil to reduce its acidity.

• Appropriate bases can neutralise acidic industrial waste before safe treatment and disposal.

6. SALTS

A salt is an ionic compound that can be formed when the hydrogen ion of an acid is replaced by a metal or ammonium ion. Many salts are produced by neutralisation of an acid with a base.

Salt	Common name / formula	Uses
Sodium carbonate	Washing soda; Na■CO■ (often supplied as Na■CO■·10H■O)	Laundry, water softening, glass and detergent industries.
Sodium hydrogencarbonate	Baking soda; NaHCO■	Baking, some antacids and certain fire extinguishers.
Potassium nitrate	KNO■	Fertilisers and some fireworks / pyrotechnic compositions.

7. QUICK FORMULA AND FACT REVISION

Name	Formula / key fact
Hydrochloric acid	HCl
Nitric acid	HNO■
Sulphuric acid	H■SO■
Phosphoric acid	H■PO■
Sodium hydroxide	NaOH
Potassium hydroxide	KOH
Calcium hydroxide	Ca(OH)■
Ammonium hydroxide (school notation)	NH■OH
Sodium carbonate	Na■CO■
Sodium hydrogencarbonate	NaHCO■
Potassium nitrate	KNO■

8. COMMON EXAM TRAPS

• Neutral solutions do not change red or blue litmus.

• Phenolphthalein is colourless in acid and neutral solution, but pink in a basic solution.

• Methyl orange is red in acid, orange near neutral, and yellow in base.

• Turmeric remains yellow in acidic and neutral solutions; it turns reddish-brown in a base.

• An alkali is a water-soluble base; insoluble bases are not alkalis.

• Acid + metal generally releases hydrogen; acid + carbonate releases carbon dioxide.

• Neutralisation forms salt and water. The salt formed depends on the acid and base used.

Safety reminder: Never taste acids or bases. Avoid direct skin contact. In a laboratory, use eye protection and follow the teacher's instructions.